THESIS DIPLOMA

DESIGN OF A HIGH TEMPERATURE HEAT PUMP

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ACKNWOLEDGMENTS

Having nearly completed my undergraduate studies in Mechanical and Manufacturing Engineering at the University of Cyprus, I am privileged to have embarked on this academic journey alongside likeminded colleagues. This endeavor has provided me with the unique opportunity to cultivate a profound understanding for the art of Engineering under the guidance of leading experts in the field.

Reflecting on this transformative experience, I extend my gratitude to my family and friends for their unweaving support throughout this odyssey. The pass was filled with challenges and trials, yet it was equally rewarding with moments, personal growth, finding new friends and the emergence of unforeseen opportunities that surpassed all my expectations.

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ABSTRACT

There is no doubt true our planet is facing one of the most serious changes modern histories has come across. Climate change because of global warming from greenhouse emissions seems to have a negative impact on our future as a human species. In recent research, data show that the number of natural disasters has dramatically increased from 291 to 904 events from 1980 to 2014 [1]. Regarding those numbers, EU has decided to take direct actions, setting up ambitious goals for the decarbonization of the planet by 2050 and decided to minimize Europe's carbon dioxide emissions by 55% until 2030 [2]. Following this enormous effort by Europe, new and efficient ways to harvest our energy resources must be examined. Heat pumps (HP) are one of the most efficient ways of transferring heat with a coefficient of performance at nearly 3-5. [3]

HP presents a promising avenue for sustainable alternatives to conventional air-conditioning systems and boilers. This assertion is predicated on the inherent capacity of heat pumps to facilitate the transfer of heat, rather than its generation, by leveraging a medium, which is called the heat pumps refrigerant. Notably the pivotal characteristic contributing to the efficiency of this technology is its ability to offer both heating and cooling functionalities, with the use of a reversing valve.

One of the most important aspects of the heat pump system is their classification as renewable energy sources, coming from their capability to harness and repurpose waste heat productively. Therefore, making heat pumps a viable instrument to harness latent energy form the environment, contributing to the sustainability of our planet. [4]

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Chapter 1

1 INTRODUCTION

1.1 Heat Pump and Existing Systems

Heat pumps represent highly efficient systems capable of transferring heat from one location to another, and their exceptional coefficient of performance underscores their status as renewable technologies. This characteristic positions them as a superior alternative to conventional thermal transfer systems. The burgeoning interest in heat pump systems has spurred extensive research efforts aimed at maximizing their potential, with a notable focus on enhancing efficiency and mitigating environmental impact.

A notable surge in research endeavors has resulted in a plethora of scholarly contributions proposing innovative strategies for advancing heat pump technology. Among these proposals, the combination of air source heat pumps, which deliver either chilled or heated air, with water source heat pumps, specializing in providing hot or cold water, has emerged as a noteworthy avenue. Research findings suggest that such hybrid systems can yield efficiency improvements of up to 20% [4]. Additionally, investigations have advocated for the integration of scroll compressors to curtail losses associated with the reversing valve [4].

Further investigations have delved into the optimization of refrigerants for heat pump systems. Notably, research has indicated that the utilization of R433A as a refrigerant can yield a commendable coefficient of performance (COP) ranging from 4.9 to 7.6 [5]. This underscores the significance of meticulous refrigerant selection in enhancing the overall performance of heat pump systems.

Also, a great amount of research in this field is involved with the introduction of combined photovoltaic systems with heat pump systems, where some such systems had concluded to a maximum COP that reaches 8.4 and an average of 6.5, whereas the average photovoltaic efficiency was just 13.4% (2010), where the photovoltaic assisted systems use the photovoltaic cells as an evaporator that provides the system with heat. Moreover, in such research the harness of waste heat water was suggested especially from saunas and pools [6].

1.2 High Temperature Heat Pumps

High temperature systems are of great interest to many researchers that try to improve the efficiency of buildings or generally find alternatives for the transition to a sustainable future [7]. Such a system can provide heat sink temperatures of 90 °C, while only 13 manufacturers can provide such systems in the industry worldwide [8]. High Temperature systems can be an innovative solution for the transition into a sustainable future since such systems can effectively be used for the purpose of handling waste heat and transforming it into different types [9].

Regarding the high-temperature heat pump design, it is noteworthy that the thermodynamic cycle remains analogous to that of lower working temperature heat pumps (below 60°C). This similarity entails the refrigerant undergoing phase changes through expansion and compression processes [10]. However, the distinctive feature of high-temperature heat pump systems lies in their requisite capacity to attain elevated temperatures, as is the focus of this study, with a target range extending up to 90 – 120 °C. The heightened temperatures involved introduce specific challenges, with electronic valve functionality and the choice of lubrication oil emerging as paramount considerations [11].

1.3 Objectives

The primary objective of this study is the conceptualization and design of a high-temperature heat pump adept at elevating water temperatures from 25 to 90 °C, while concurrently possessing the capability to cool water down to 10 °C. The efficacy of such a design hinge on the meticulous definition of various parameters to maximize system efficiency. Key parameters include the determination of the refrigerant's mass flow rate, the initial and maximum pressure parameters within the system, and the establishment of the working temperatures for both the heat-up process and the chosen refrigerant. These parameters collectively constitute crucial facets that necessitate precise definition to realize the optimal efficiency of the high-temperature heat pump system under consideration.

The goal of this study, will be the identification of all the critical parameters involved in a Heat Pump system, trying to understand them and manipulate them in a manner that can create the most efficient system possible, taking into consideration limitations that might exist. For this analysis a mathematical model is created based on a single-stage cycle, while also considering a steady state analysis, without taking into consideration any dynamic behavior.

Chapter 2

2 THEORETICAL BACKGROUND AND MODEL

2.1 Theory for the thermodynamic cycle of a Heat Pump

The thermodynamic cycle of a heat pump system comprises four integral components, as illustrated in Figure 2.1. The initial component of this system is the compressor, situated between points 1 and 2. While the process within the compressor is characterized by constant entropy, a more comprehensive model necessitates the incorporation of a coefficient to account for the isentropic expansion of the compressor, rendering the model more representative of real-world conditions. The second and fourth components are the heat exchangers, pivotal in enhancing the system's efficiency by facilitating the transfer of heat between different mediums. This heat transfer occurs at constant pressure, effecting the transformation of the refrigerant from its vapor state into a fully fluid state. Conclusively, the fourth component is the expansion valve, tasked with the refrigerant's expansion, thereby transforming it from a fluid state to a saturated mixture. Theoretically, the process within the expansion valve is isenthalpic.

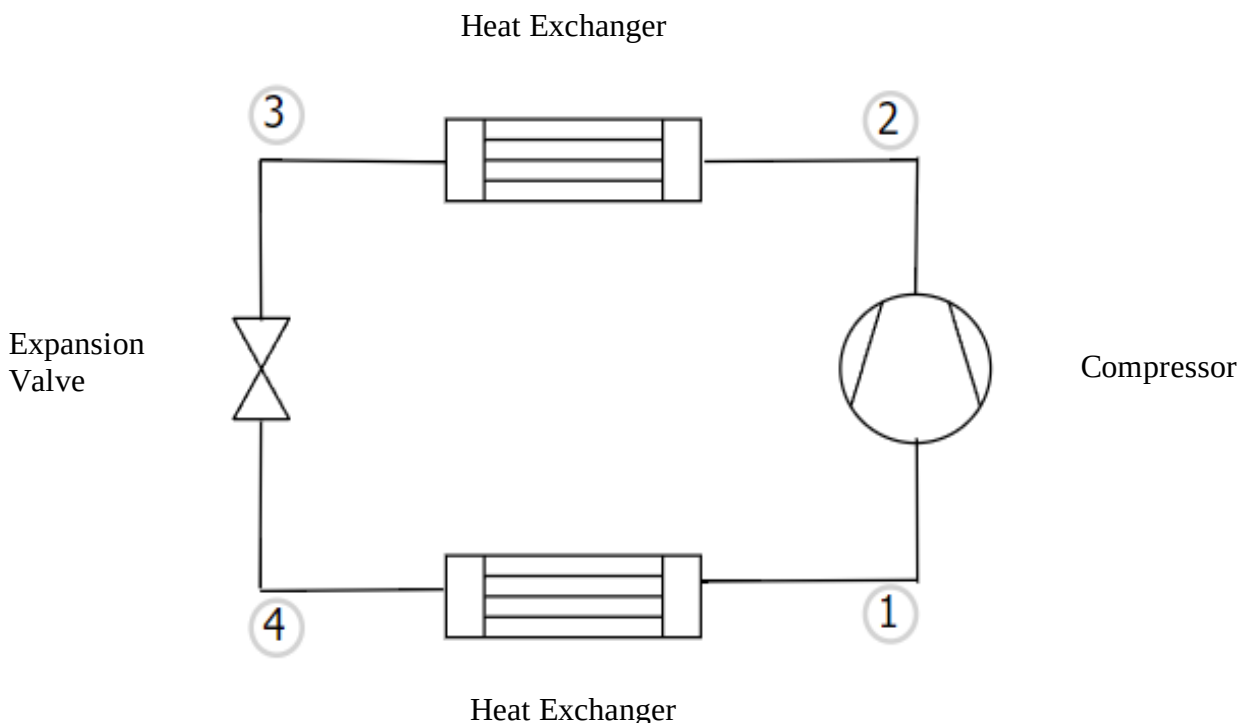

Figure 2.1: Heat Pump System Components.

2.2 Basic Model

The analysis of the systems requires a steady state analysis of a vapor-compressor cycle, while every element of the heat pump will be studied separately making the required assumptions.

For the analysis of the different components of the system the first thermodynamic law will be applied (Conservation of Energy) [12].The analysis is based on a self-developed code in MatLab by importing coolprop library guided by python (Appendix I) [13].

2.2.1 Assumptions

1) The flow of the water outside the heat exchanger is considered homogenous.
2) The flow of the fluid passing through the heat exchanger is considered homogenous.
3) Steady state conditions
4) No energy losses in the total system
5) No heat losses from the heat exchanger
6) Heat transfer rate of a pipe's refrigerant will be simultaneous.

The above assumptions are introduced with the intent of streamlining the mathematical formulations, justified by the expectation that these simplifications will not substantially deviate from the anticipated results. Nevertheless, it is imperative to acknowledge that the final model will be refined to incorporate a coefficient that encapsulates the losses inherent in the system, drawing upon empirical evidence to enhance its accuracy. This coefficient will serve to systematically account for and aggregate the various losses incurred within the heat pump system.

It is crucial to underscore that the proposed model represents an inaugural step in a broader research trajectory, with its primary function centered on facilitating comparative analyses among diverse working fluids, specifically different refrigerants utilized in heat pump systems. This model is poised to be a valuable tool for comprehending the intricate interplay of various parameters within the system. Specifically, it is envisaged that the model will serve to estimate preliminary values for these parameters, providing a foundational framework for subsequent, more detailed investigations.

In essence, the current model serves as a strategic precursor, laying the groundwork for future endeavors by offering a comprehensive understanding of the system's intricacies. Its application is oriented towards discerning disparities among heat pump working fluids, thereby contributing to the systematic advancement of knowledge in this domain.

2.2.2 Heat Pump's System Components Mathematical Modelling

The analysis in this section, refers to the four working stages of a heat pump as shown previously in fig 2.1

Stage 1-2: Compressor

Assumptions made for the mathematical modeling of the compressor:

- Isentropic procedure
- The Isentropic pump's efficiency of n = 0.85
- Refrigerant flow rate will vary

The maximum pressure of the system is set at 1.5 MPa while the initial pressure varies in respect to the different models that will be examined. The temperature, entropy and enthalpy of point 1 (before the compressor) and point 2 (after the compressor) will be given by TPSI tables that were introduced by the addition of the coolprop library. To fully define a thermodynamic stage, the program must be provided with at least two thermodynamic properties. For stage one the second thermodynamic property is the quality of the mixture, which is exactly 1 (the mixture is at fully vapor stage) [14]. By identifying, entropy (s_1) , enthalpy(h_1), and temperature (T_1)at point 1, the process can go on for point 2 since the process is isentropic (entropy at point 1 is equal to the one at point 2). From the 1st law of Thermodynamics, it follows that the isentropic work done by the compressor is,

$$W_{compressor-s} = h_{2s} - h_1 \quad (1)$$

where s symbolizes isentropic enthalpy. The actual work done by the compressor is,

$$W_{compressor} = \frac{W_{compressor-s}}{n} \text{ [J/kg]} \quad (2)$$

$$h_2 = W_{compressor} + h_1 \quad (3)$$

For the calculation of the compressors power, the following equation is used

$$P_{compressor} = W_{compressor} \times \dot{m}_{refrigerant} \text{ [W]} \quad (4)$$

where $\dot{m}_{refrigerant}$ is the mass flow rate of the refrigerant [kg/s], and $P_{compressor}$ is the power [W] required to drive the compressor.

Due to the inability of the coolprop library to provide values for pressure with inputs of temperature and enthalpy further analysis will be continued using the isentropic values of pressure.

Stage 2-3 and 4-1: Heat Exchanger

For stage 2 to 3 the Heat Exchanger acts as a condenser while at stage 4 to 1 it acts as an evaporator.

The basic assumptions made for the mathematical modelling of the Heat Exchangers include:

- Initial fluid temperature of 25 $°C$
- Mass flow rate of the outside fluid will vary
- All the heat energy of the refrigerant will be transferred to the fluid.

Heat exchangers represent devices designed to facilitate the transfer of heat from a fluid at a higher temperature to a fluid at a lower temperature. Under the assumption that the quantities of desorption and absorption are equivalent for the two fluids involved, it becomes feasible to derive Equation 2 through the utilization of Equation 1 [15]. This assumption implies a balanced exchange of thermal energy between the fluids undergoing desorption and absorption processes, thereby establishing a foundational relationship encapsulated by the equations. It must be noted that n = 1 for a heat exchanger and therefore the above equations can be used for the derivation of equation 6 and 7.

In this context, equation 6 serves as a critical starting point, presumably outlining the governing principles of the heat exchanger operation. Through careful consideration of the equality between desorption and absorption, equation 6 is deduced, providing a quantitative expression that encapsulates the dynamic equilibrium of heat transfer between the higher and lower temperature fluids within the heat exchanger system.

This fundamental relationship, as elucidated by the derived equations, lays the groundwork for a deeper understanding of the thermal dynamics inherent in heat exchangers operating under the specified conditions. The equality assumption regarding desorption and absorption forms a pivotal basis for the subsequent mathematical representation, contributing to the elucidation of heat transfer mechanisms in such systems.

Due to thermodynamic cycle, pressure at the points before and after the heat exchanger must be the same, thus providing the equation:

$$p_2 = p_3 \quad (5)$$

Moreover, the quality of the mixture at stage 3 must be equal to zero since it will be at fluid stage.

Using those two known parameters data for the enthalpy, entropy and Temperature can be extracted.

Then the heat Transfer rate exchanged from the refrigerant to the fluid can be extracted using equations 6 and 7. [28]

$$Q_{23} = \dot{m}\,(h_3 - h_2)\ [\mathrm{W}] \quad (6)$$

$$Q_{41} = \dot{m}\,(h_1 - h_4)\ [\mathrm{W}] \quad (7)$$

Stage 3-4: Expansion valve

Due to theory of isenthalpic expansion the enthalpy at point 4 is equal to enthalpy at point 3 and pressure at point 4 is equal to pressure at point 1, thus,

$$h_3 = h_4 \quad (8)$$

Coefficient of Performance

The coefficient of performance is the most important factor for the realization of efficiency of the system. It represents the amount of thermal energy produced against the electrical energy that must be supplied to the compressor. The highest value for the COP indicates higher efficiency and therefore optimal operation. [16]

$$COP = \frac{Q_{23}}{P_{compressor}} \quad (9)$$

2.3 Expanded analysis

All the models presented in this section are based on the Basic Model presented in section 2.2.

2.3.1 Mass Flow Rate Analysis

The mass flow rate model is developed to examine the behavior of the refrigerant's mass flow rate and to gain insights into the impact that alterations in this parameter exert in the efficiency of the model.

This model employs identical equations as the basic model, with the notable inclusion of a for loop. This loop iterates through various values of the mass flow rate, offering a comprehensive examination of the compressor's power and the heat transfer rate within the two heat exchangers. Furthermore, the model provides results regarding the system's coefficient of performance in both heating and cooling modes, allowing for a comparative analysis against different values of this parameter (Appendix II).

2.3.2 Initial Pressure

The initial pressure model is crafted with the aim of determining the optimal value for the system's initial pressure, denoted as P1.

This model closely mirrors the mass flow rate model, with a distinctive alteration which is that instead of employing a for loop to vary the mass flow rate, it employs a for loop to systematically vary the initial pressure. All other parameters are maintained at constant values, including the refrigerant type, mass flow rate, and the maximum pressure (P2), which is fixed at 1.5 MPa (a standard working pressure for the majority of commercial Heat Pumps).

This model generates graphical representations of the thermodynamic cycle, illustrating variations in entropy, enthalpy, pressure, and temperature. Additionally, it furnishes graphical depictions of the COP in both heating and cooling modes, the power consumption of the heat pump, and the heat exchange rates of the heat exchangers. All graphical results are presented in relation to the initial pressure of the system. The significance of this model lies in its pivotal role in the Heat Pump System design process, as the determination of the initial pressure enables the identification of temperatures throughout the system, thereby facilitating the selection of an appropriate refrigerant (Appendix II).

2.3.3 Refrigerant Selection

The overarching goal of the refrigerant model is to conduct a comprehensive comparison among various refrigerants. The primary objective is to identify the refrigerant that can offer the highest Coefficient of Performance (COP) in both heating and cooling modes, along with maximizing the heat exchange rates and, ideally, minimizing the power consumption of the compressor. This model serves as a critical tool for evaluating and selecting the most efficient and effective refrigerant for the given heat pump system (Appendix II).

2.3.4 Initial Pressure Variation for R717

The selection of R717 as the designated refrigerant, a reevaluation of the optimal initial pressure is imperative. The forthcoming analysis will focus exclusively on deriving fitted curves for critical parameters, including the heat exchange rates for the condenser and evaporator, alongside the power consumption of the compressor. This refined assessment aims to identify the most suitable initial pressure for the application of R717 within the specified heat pump system, ensuring an optimized and efficient operational configuration.

2.3.5 Heat Losses Introduction

The newly devised model incorporates considerations for losses by implementing a reduction in pump efficiency, characterized by a coefficient of 0.8. Additionally, the model introduces a lowered efficiency parameter of 0.8 for the heat exchangers. Operating under a set of constant parameters which are an initial pressure of 300 kPa, a refrigerant mass flow rate fixed at 0.08 kg/s, a maximum pressure of 1.5 MPa, and utilizing the standard refrigerant R717. The model is poised to provide a more realistic representation by accounting for efficiency reductions in both the pump and the heat exchangers (Appendix III).

2.3.6 Introduction of a Solar Thermal System

2.3.6.1 Mathematical Analysis of a Solar Thermal System

This model has been conceived with the primary objective of enhancing the overall efficiency of the heat pump system. This efficiency augmentation is envisioned through the integration of a solar thermal system, designed to supply pre-heated water to the pump. This strategic inclusion aims to mitigate the system's reliance on high-energy consumption, concurrently reducing the required maximum power needed from the compressor.

The theoretical underpinning of this enhancement involves the incorporation of a passive solar thermal system employing a covered absorber. The system is envisioned to operate at an average temperature of 32°C, considering the climatic conditions prevalent in Cyprus. The solar power incident on the system, denoted as G, is approximated at 800 W/m^2.

In the theoretical framework, the efficiency of any solar thermal system is contingent upon two key constants: the coefficient of refraction of the cover (τ_{cov}) and the coefficient of absorption of the absorber (a_{abs}), other parameters responsible for the thermal solar system efficiency is the environments temperature (T_∞) the temperature of the panel itself (T) and the area of the panels (A). These constants establish the upper limit of the system's efficiency. However, it is crucial to acknowledge that additional reductions in system performance may transpire, attributed to conduction and convection effects. This comprehensive theoretical groundwork lays the foundation for evaluating the solar thermal system's potential impact on the heat pump's efficiency.

$$Q = (\tau_{cov}\, a_{abs} A G) - [UA\,(T - T_\infty)] - [\varepsilon\, \sigma\, A\,(T^4 - T_\infty^4)] \ \ [\mathrm{W}] \quad (10)$$

Regarding equation 10, the term $[UA\,(T - T_\infty)]$ are the thermal losses due to conduction and $[\varepsilon\, \sigma\, A\,(T^4 - T_\infty^4)]$ are the thermal losses due to radiation, with ε and σ being the black body coefficient of the absorber and Stefan Boltzmann constant respectively. T is the temperature of the absorber and T_∞ is the temperature of the environment and A is the Area of the cover. Considering equation 10 and the associated theory, a reasoned efficiency for the solar thermal system is deemed to be a minimum of 0.40 (n). This stipulated efficiency level represents a logical threshold, reflecting the expectations and theoretical considerations within the context of the solar thermal system's performance. With the total power provided by the solar thermal system being extracted by equation 11.

$$P = G\, n\, A \qquad (11)$$

2.3.6.2 Mathematical Analysis for the Introduction of the Solar Thermal System in the Heat Pump

Assumptions made for the mathematical modelling of the thermal solar system when combined with the Heat pump system:

- The process of the refrigerant heating in the solar-thermal system is isentropic.
- In this model thermodynamic equations and laws will be used.

In contrast to prior models, the current iteration incorporates an additional stage preceding the pump. This newly introduced stage entails the integration of the heated refrigerant sourced from the thermal solar system. This augmentation represents a significant departure from the earlier configurations, signifying a more intricate and multifaceted system architecture.

The new stage will take place in between stage 1 and 2 as shown in figure 2.2 The equations for the energy provided by the absorber as well with the mass Equilibrium at the solar thermal system (st) and the energy equilibrium at the solar thermal system are given by the equations 12 , 13 and 14 respectively.

$$Q = \frac{P}{mass\ flow\ rate\ of\ refrigerant} \text{ [J/kg]} \quad (12)$$

$$m_1 = m_{st} \quad (13)$$

$$h_{st} = h_1 + Q \quad (14)$$

Utilizing the known enthalpy values and the provided entropy information, it becomes feasible to determine the pressure at the solar thermal stage. This process involves leveraging the thermodynamic properties of the system to derive the corresponding pressure values based on the available enthalpy and entropy data.

After the above calculations, h_{st} takes the place of h_1 in the basic model (Appendix IV).

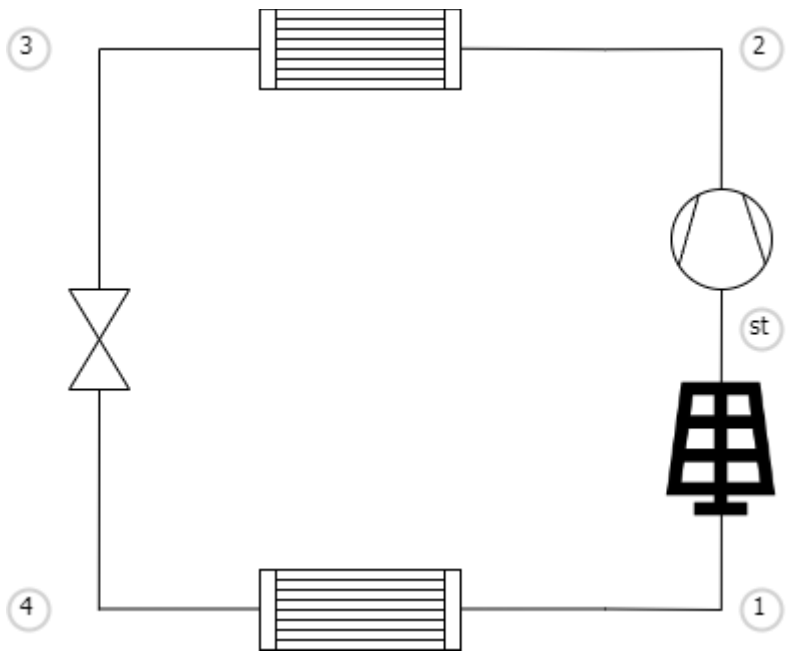

Figure 2.2: Introduction of Solar Thermal in the Heat Pump Array

Chapter 3

3 RESULTS

3.1 Mass Flow Rate Results

For the mass flow rate model, the maximum pressure of the system (P2) is set at a constant value of 1.5 MPa, while the initial pressure was set at 50 kPa, while also having set a constant refrigerant to be R152a. Using the model provided data for all the thermodynamic properties of the system at the four distinct points of interest.

From Equation 9 anticipated outcome for the Coefficient of Performance (COP) is its resilience to fluctuations in the mass flow rate of the refrigerants. This expectation stems from the fundamental nature of the COP calculation, where the division of the thermal power at points 2 and 3 (Q23) by the power of the compressor (W_compressor) results in the cancellation of the mass flow rate, rendering it inconsequential to the final COP value. Consequently, variations in the mass flow rate are not expected to influence the COP, affirming its stability in the face of changes in this specific parameter.

In fig 3.1 validation is substantiated by the graphical representation of the Coefficient of Performance (COP), which exhibits a consistent and unvarying trend in response to changes in the mass flow rate. The COP graph manifests a noteworthy stability, affirming the theoretical expectation that the COP remains unaffected by alterations in the mass flow rate. This empirical observation lends support to the underlying principles of the thermodynamic model, reinforcing the notion that the division of thermal power and compressor power in the COP calculation effectively mitigates the influence of mass flow rate variations on the final COP values.

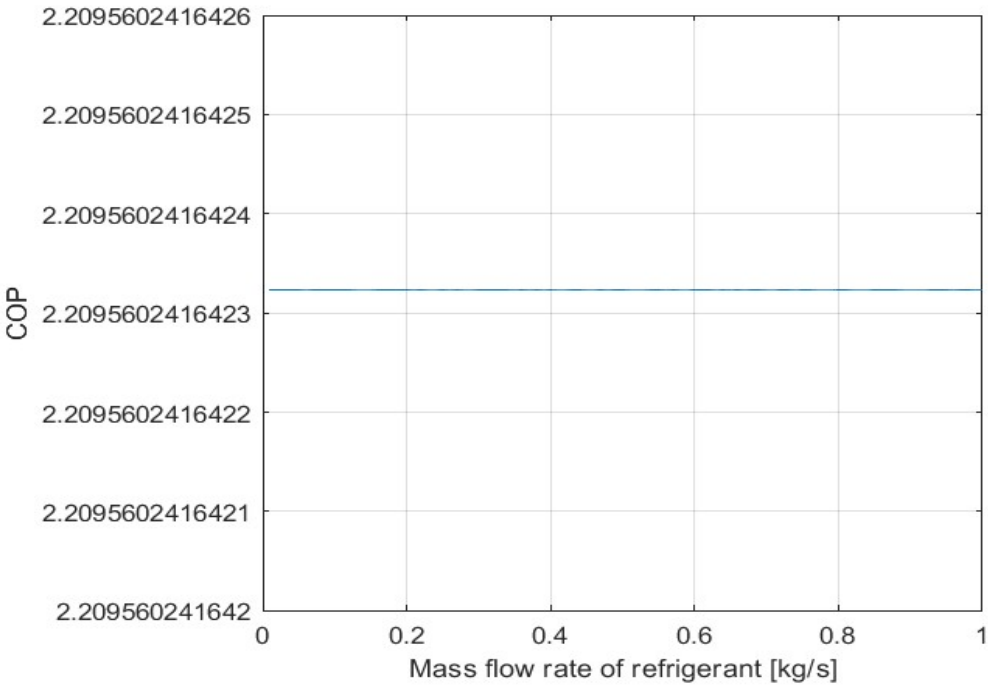

Figure 3.1: COP in heating mode against mass flow rate of the refrigerant.

Equations 4, 6 and 7, shows that mass flow rate of the refrigerant in proportional to the values assigned to the heat transferred rate and the power of the compressor.

The graphical representation of fig 3.2, 3.3 and 3.4 clearly demonstrates a proportional relationship between the mass flow rate of the refrigerant and the assigned values for both the heat transfer rate and the power of the compressor. This observed correlation underscores the interdependence of these parameters within the system, as changes in the mass flow rate manifest concomitant effects on both the heat transfer rate and compressor power. The graphical evidence reinforces the inherent connection between these key variables, providing valuable insights into the behavior of the heat pump system, while the results are expected because of equations 4, 6 and 7. However the importance of those graphical representations lies in the values in kW for Q23, Q41 and $P_{compressor}$.

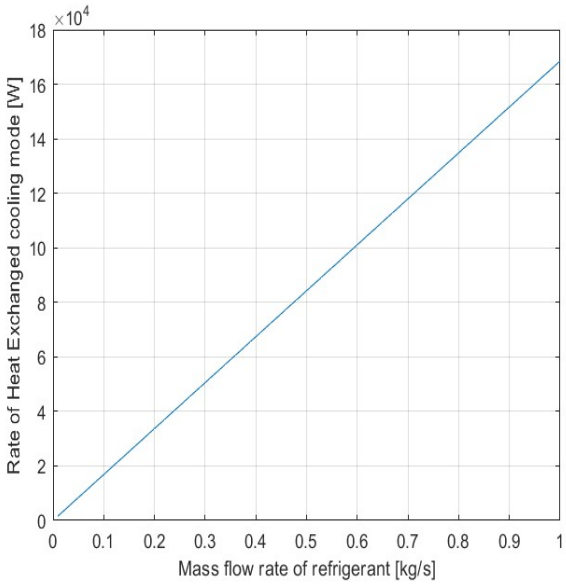

Figure 3.2: Heat Exchanged Rate for Cooling mode for the mass flow rate results.

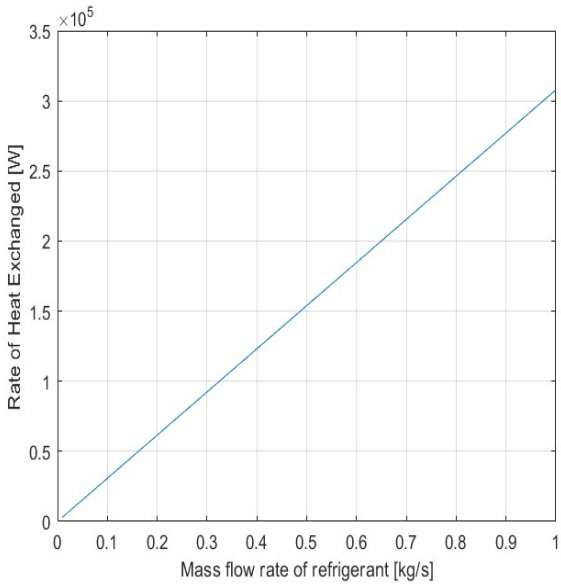

Figure 3.3: Heat Exchanged rate for Heating mode for the mass flow rate results.

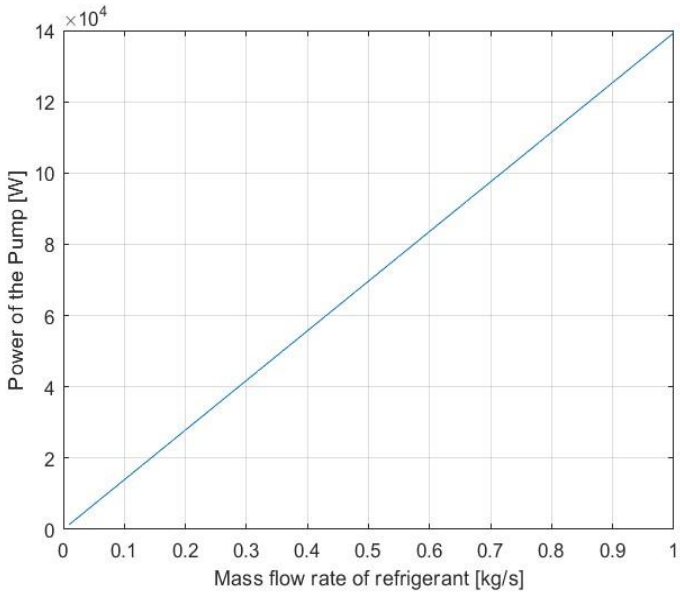

Figure 3.4: Power for the Compressor for the mass flow rate results.

The obtained results pertaining to the thermodynamic properties at the four distinct points of the cycle align with expectations, conforming to the prescribed thermodynamic cycle established within the model. These outcomes are visually depicted in figure 3.5, providing a graphical representation that illustrates the adherence of the system to the specified thermodynamic cycle.

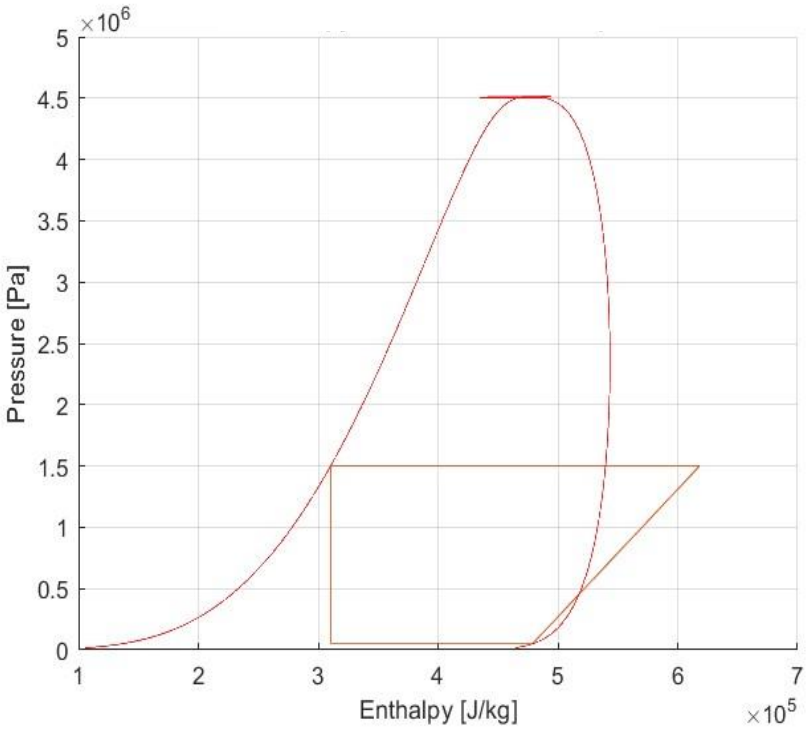

Figure 3.5: Pressure against Enthalpy Graph for mass flow rate of the refrigerant being 0.06 kg/s.

The assigned value for the mass flow rate of the refrigerant is strategically set within the range of 0.06 to 0.1 kg/s, aiming to minimize the power consumption of the heat pump. This optimization objective arises from the observation of a compressor power (W compressor) equal to 140 kW in the initial results. Throughout the course of the paper, this mass flow rate value will be subject to variation as part of the ongoing efforts to identify a more optimal and suitable parameter configuration. The iterative adjustment of the mass flow rate is motivated by the pursuit of enhanced system performance in consideration of evolving parameters selected during the analytical process.

3.2 Initial Pressure

The initial Pressure Model has the purpose of assigning a value to this parameter. The model will be running for values for the initial pressure starting from 30 to 200 kPa while having a step of 50 kPa, while setting constant values for the mass flow rate of the refrigerant at 0.06 kg/s, the maximum pressure as before at 1.5 MPa and as the previous model the refrigerant to be R152a.

Results for pressure against enthalpy are as expected following the thermodynamic cycle already set in the model and are shown in figure 3.6.

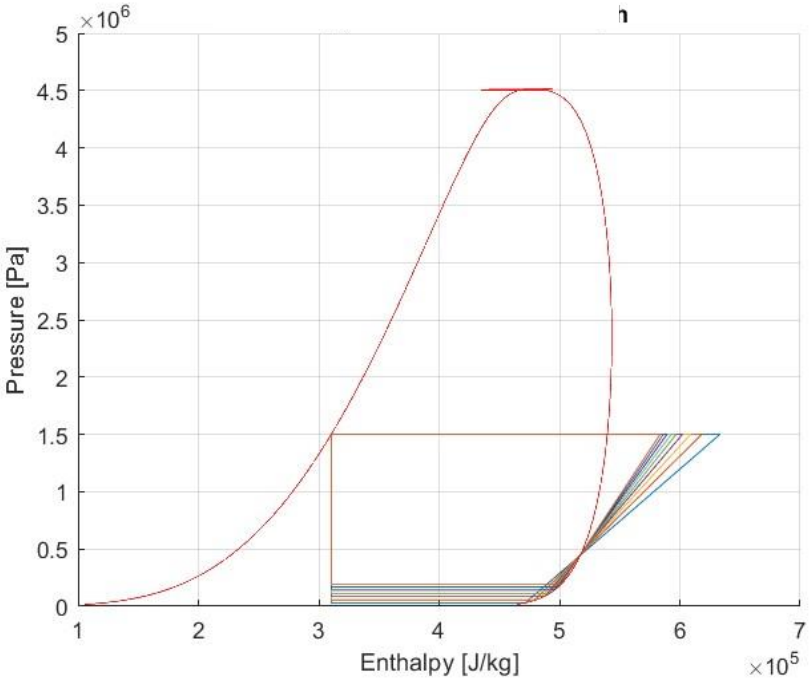

Figure 3.6: Pressure against Enthalpy with varying initial pressure.

The examination of an increased initial pressure in the heat pump system reveals a discernible upward trend in the Coefficient of Performance (COP) for both heating and cooling modes, as elucidated in Figures 3.7 and 3.8. The results exhibit an almost proportional relationship between the pressure values and the COP, indicating that the COP tends to rise with higher initial pressure values. Consequently, the optimal selection for the initial pressure should lean towards higher values. However, it is crucial to balance this consideration with the temperatures observed at the four key points of the system, ensuring that the chosen initial pressure aligns with the thermal requirements of the heat pump and the desired efficiency levels.

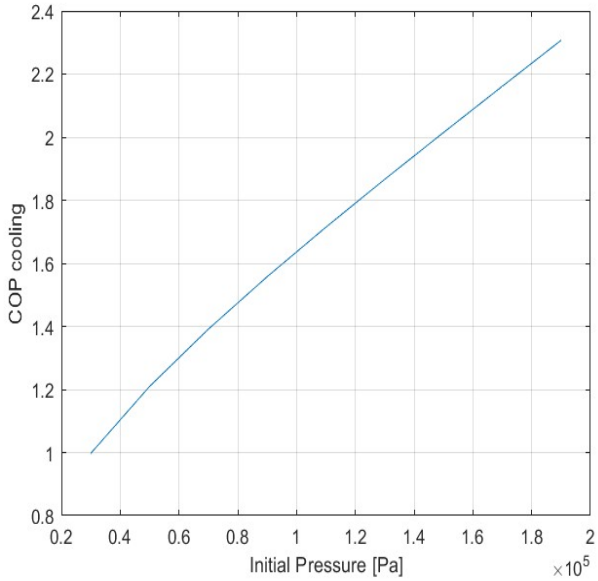

Figure 3.7: COP cooling mode with varying initial pressure.

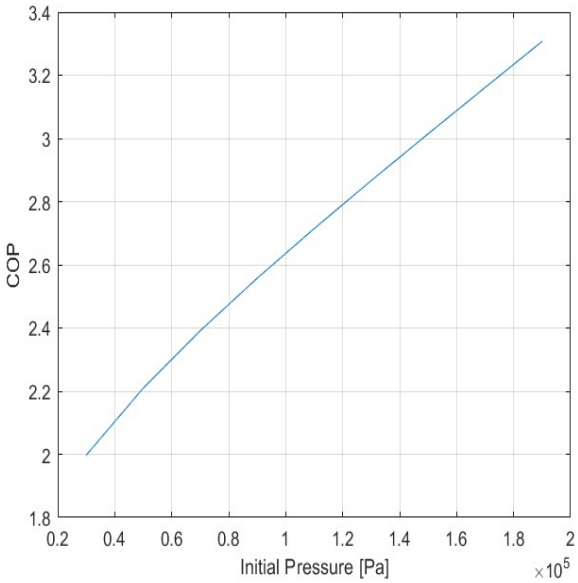

Figure 3.8: COP heating mode with varying initial pressure.

As depicted in Figures 3.9 and 3.10, there is a noticeable decline in both the rate of heat exchange for the condenser and the power consumption of the compressor with an increase in the initial pressure of the system. In contrast, figure 3.11 illustrates a concurrent rise in the heat exchange rate of the evaporator. These findings emphasize the intricate trade-off that needs to be navigated in determining the optimal initial pressure for the system.

The observed results suggest that the ideal initial pressure should strike a delicate balance, aiming for both high heat exchange rates in both cooling and heating modes while simultaneously minimizing the power consumption of the heat pump. This intricate

equilibrium is crucial for achieving optimal system efficiency. It is worth noting that this optimization process is influenced not only by the initial pressure but also by other system parameters, including the mass flow rate of the refrigerant, as discussed in Section 3.1. The interplay of these factors necessitates a comprehensive approach to system design, considering multiple variables for achieving the desired performance outcomes.

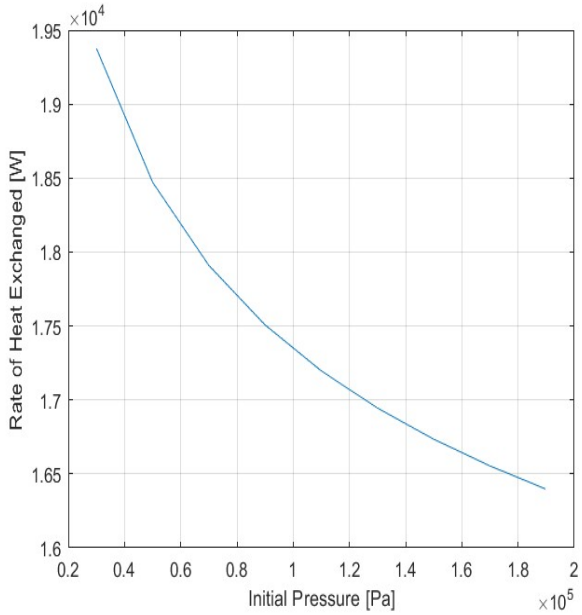

Figure 3.9: Rate of Heat Exchange Heating mode with varying initial pressure.

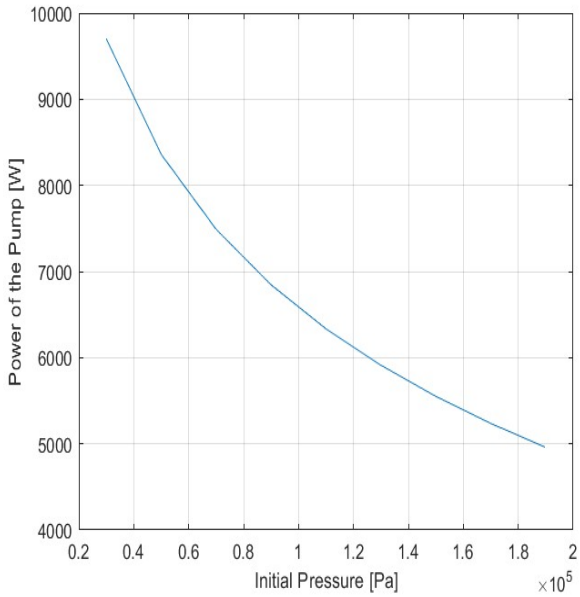

Figure 3.10: Power Consumption of Compressor with varying initial pressure.

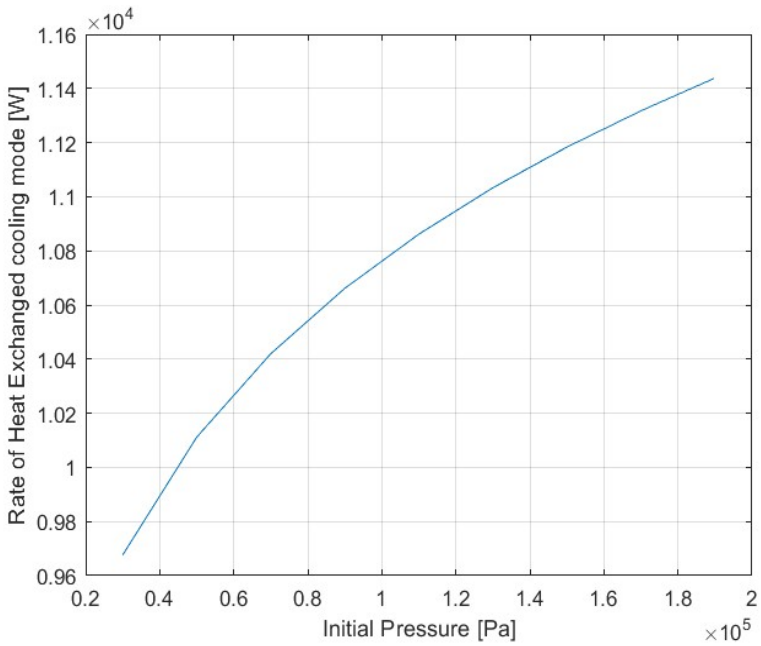

Figure 3.11: Rate of Heat Exchanged in Cooling mode with varying initial pressure.

The results for the heat exchanged and the power of the compressor will be fitted following a $y=x^4$ trend using MATLAB's polyval and polyfit function in such a way. The results are shown in figures 3.12, 3.13 and 3.14.

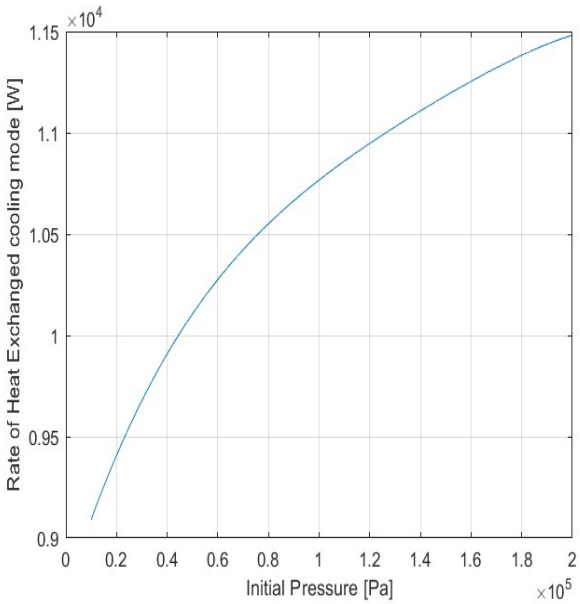

Figure 3.12: Heat exchanged rate heating mode fitted in x^4.

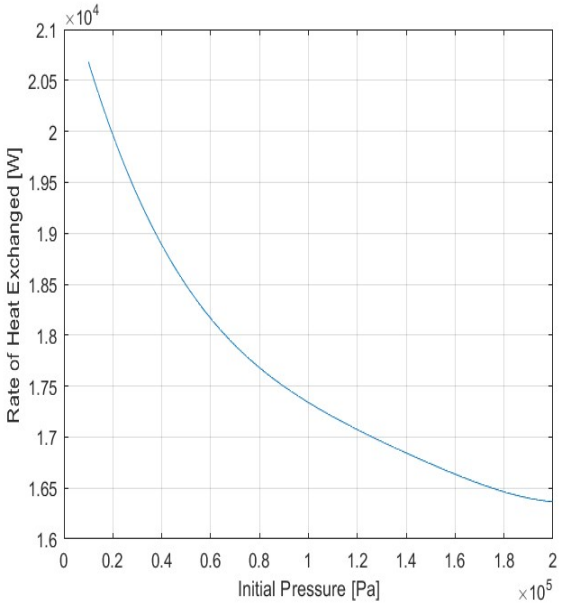

Figure 3.13: Heat exchanged rate cooling mode fitted in x^4.

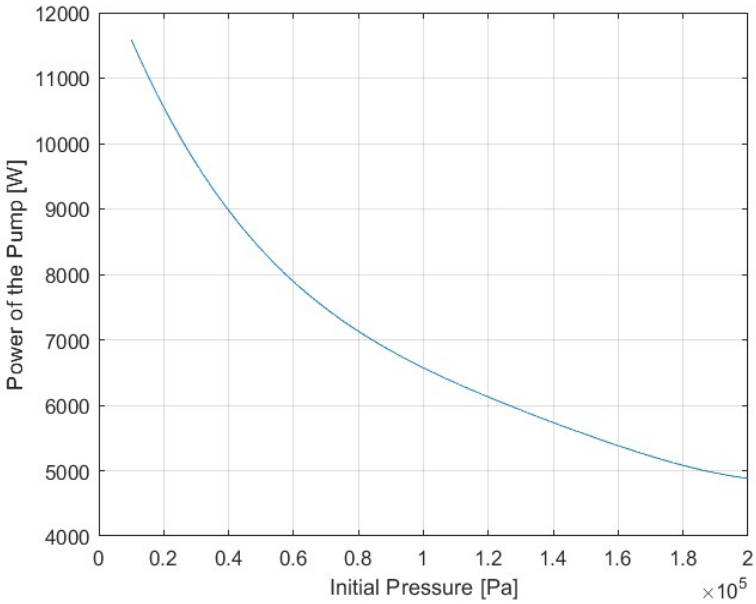

Figure 3.14: Power of the Compressor fitted in x^4.

Polynomial results

Equations 15, 16, and 17 represent the outcomes of the polynomials derived using MATLAB's built-in functions, specifically polyfit and polyval. These equations encapsulate the mathematical expressions characterizing the relationships associated with the heat exchange rate in both cooling and heating modes, along with the power consumption of the heat pump, respectively. These polynomial representations serve as valuable tools for predictive modeling and analysis, offering a concise and functional means of describing the system's behavior in response to varying parameters.

$$Q_{23} = (-2.3274 \times 10^{-18})P^4 + (1.3356 \times 10^{-12})P^3 - (3.0423 \times 10^{-7})P^2 + 0.0398P + (8.7224 \times 10^3) \quad (15)$$

$$Q_{41} = (-6.9609 \times 10^{-18})P^4 - (3.9020 \times 10^{-12})P^3 - (8.4499 \times 10^{-7})P^2 - 0.0945P + (2.1546 \times 10^4) \quad (16)$$

$$Compressor = (-9.2883 \times 10^{-18})P^4 - (5.2375 \times 10^{-12})P^3 + (1.1492 \times 10^{-6})P^2 - 0.1344P + (1.2824 \times 10^4) \quad (17)$$

Based on the findings from the initial Pressure Model, the data indicates that the optimal performance for the heat pump system is achieved within the initial pressure range of 200 kPa to 300 kPa. This conclusion holds while considering a maximum pressure of 1.5 MPa and a mass flow rate of the refrigerant ranging from 0.06 kg/s to 0.08 kg/s. These specific parameter

values, falling within the mentioned ranges, demonstrate superior heat pump efficiency and performance.

3.3 Refrigerant Selection

The model will operate under consistent conditions, maintaining fixed parameters throughout its execution. Specifically, the model will utilize an initial pressure of 50 kPa and 200 kPa, a maximum pressure capped at 1.5 MPa, and a constant mass flow rate of 0.08 kg/s. These stable parameters ensure a controlled environment for the comparative assessment of different refrigerants.

The initial run was conducted at an initial pressure of 50 kPa, and the results for the Coefficient of Performance (COP) in both heating and cooling modes for the various refrigerants are presented in Table 3.1. The designation "NA" indicates instances where the assigned pressure values were not compatible with the thermodynamic properties of the respective refrigerants.

In Table 3.2, a subsequent test was executed at an initial pressure of 200 kPa. This table provides data for the highest temperature within the system, the heat exchange rate at the condenser, and the power consumption of the compressor for the same set of refrigerants. These results offer insights into the performance of each refrigerant under altered pressure conditions, allowing for a comparative analysis of their thermodynamic behavior.

Table 3.1: COP for different refrigerants [17] for an initial Pressure of 50 kPa

Refrigerant	COP Cooling	COP Heating
R152a	1.2096	2.2096
R12	0.9925	1.9925
R144a	(NA)	(NA)
R717 (Ammonia)	1.5378	2.5378
R744	(NA)	(NA)
R134a	1.1124	2.1124
R1234yf	0.7704	1.7704
R410a	1.3254	2.3254
R407c	1.2916	2.2916
R290 (propane)	0.9906	1.9906
R600 (Butane)	0.6174	1.6174

R600a (isobutane)	0.6195	1.6195

Table 3.2: Heat exchanged rate of Condenser and Power Consumption of the Heat Pump for an initial Pressure of 200 kPa

Refrigerant	T2 [°C]	Condenser [kW]	Power Compressor [kW]
R152a	80	21.77	6.440
R12	69	10.39	3.40
R144a	(NA)	(NA)	
R717 (Ammonia)	129	112.96	28.57
R744	(NA)	(NA)	
R134a	62	12.98	3.959
R1234yf	56	9.48	3.43
R410a	53	19.26	5.43
R407c	59	16.51	4.70
R290 (propane)	54	27.11	8.95
R600 (Butane)	99	20.19	8.047
R600a (isobutane)	75	19.09	7.65

The analysis of Tables 3.1 and 3.2 reveals that R717 (Ammonia) emerges as the refrigerant with the highest Coefficient of Performance (COP) for both heating and cooling modes. Additionally, R717 exhibits the highest heat exchange rate for its condenser. However, it is noteworthy that the compressor's power consumption for R717 is comparatively higher than that of other refrigerants, although still within reasonable limits.

It is crucial to acknowledge that R717 is categorized as a flammable and toxic refrigerant [18]. Therefore, stringent safety measures must be implemented to ensure the well-being of individuals in proximity to the system.

3.4 Refrigerant Characteristics

In this updated model configuration, R717 will be designated as the fixed refrigerant. The mass flow rate will be held constant at 0.08 kg/s, and the maximum pressure will remain capped at 1.5 MPa. The curve fitting procedure will involve the use of third-order polynomials, chosen for their simplicity and effectiveness in capturing the trends of the heat exchange rates for the condenser and evaporator, as well as the power consumption of the compressor.

The outcomes subsequent to the fitting of the data have been visually represented in Figures 3.15 , 3.16 and 3.17. Meanwhile, the mathematical expressions encapsulating the fitted trends in the form of third order polynomials are elucidated in Equations 18, 19, and 20 which describe the Heat Exchange rate for the cooling mode of the heat Pump, the Heat exchange rate for the Heating mode as well as Power consumption of the compressor respectively. These graphical and analytical representations serve to distill and convey the relationships derived from the data fitting process, providing a structured and informative presentation of the results.

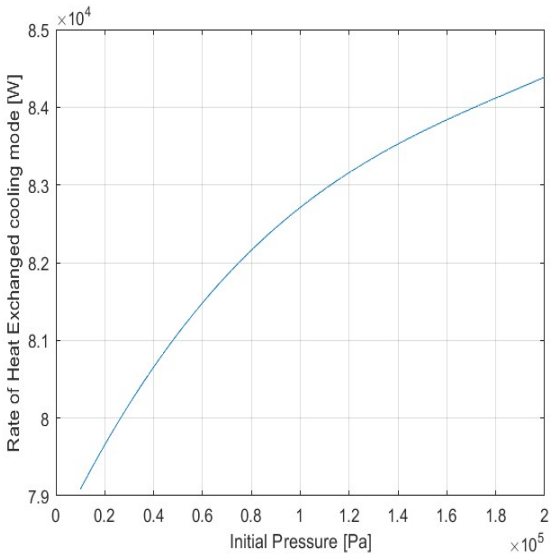

Figure 3.15: Rate of Heat Exchanged Cooling Mode fitted in x^2 for R717 with varying pressure.

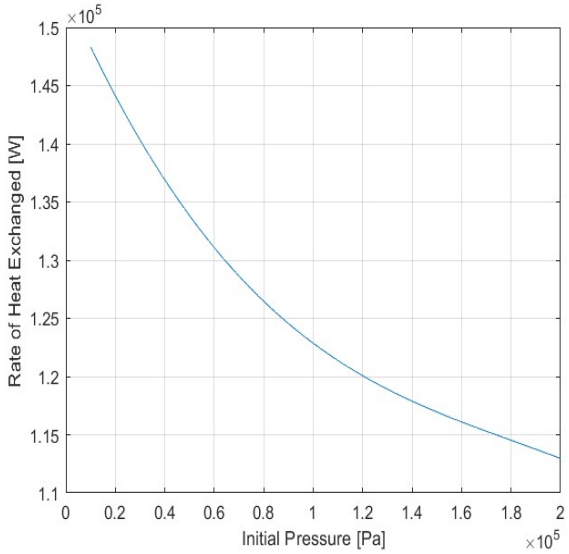

Figure 3.16: Rate of Heat Exchanged Heating Mode fitted in x^2 for R717 with varying pressure.

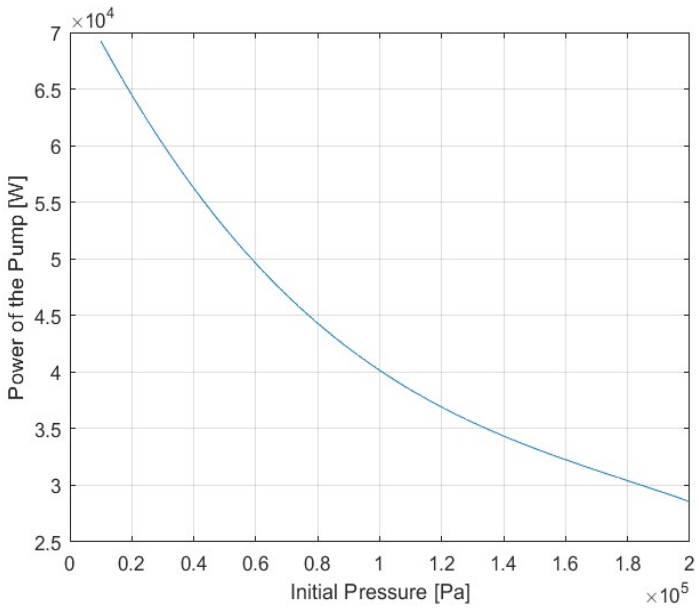

Figure 3.17: Power Consumption of The Compressor fitted in x^2 for R717 with varying pressure.

$$Q_{23} = (3.8892 \times 10^{-7})P^2 - 0.2358P + (1.4384 \times 10^5) \quad (18)$$

$$Q_{41} = (-5.4638 \times 10^{-8})P^2 + 0.0355P + (7.9546 \times 10^4) \quad (19)$$

$$W_{compressor} = (4.4456 \times 10^{-7})P^2 - 0.2714P + (6.4296 \times 10^4) \quad (20)$$

Using extreme value theorem [19] equation 18 must be differentiated and then set equal to zero to find the value of initial pressure that the maximum value for the Rate of Heat

Exchanged for cooling mode can be achieved. The value for this pressure is therefore equal to 302 kPa.

Figures 3.18 and 3.19 show the COP for heating and cooling respectively.

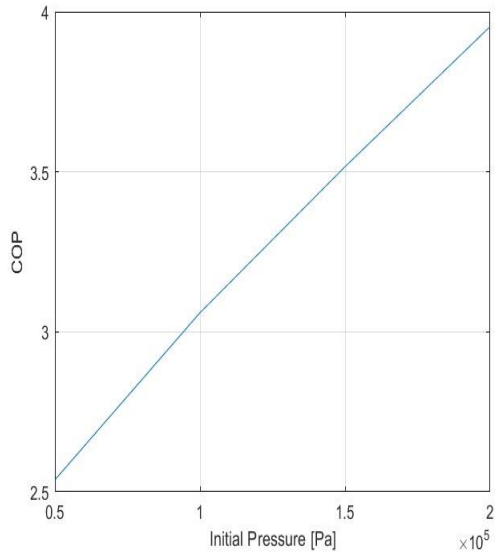

Figure 3.18: COP Heating Mode for R717 with varying pressure.

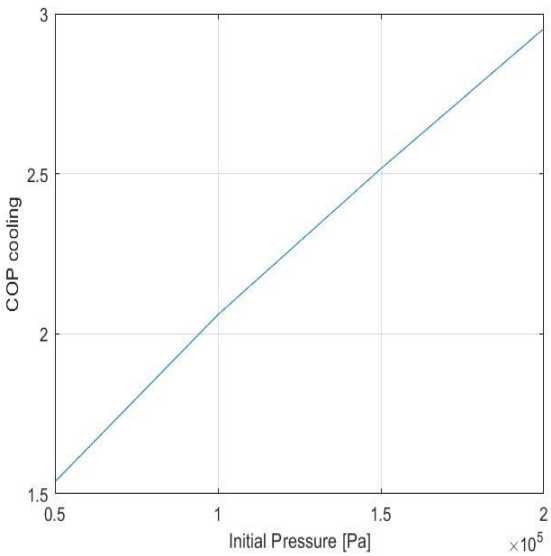

Figure 3.19: COP Cooling Mode for R717 with varying pressure.

In the process of determining the optimal initial pressure that can yield the required levels of both heating and cooling rates, it is imperative to acknowledge the intricacies inherent in the problem. The deployment of a high-temperature heat pump with an emphasis on heightened efficiency necessitates the attainment of maximal heat exchange rates coupled with minimized compressor power consumption. Considering this consideration and considering the notably

high heat exchange rate achievable with R717, a judicious selection has been made to set the initial pressure at 300 kPa. This choice yields a Coefficient of Performance (COP) for heating of 4.84 and for cooling of 3.84, aligning with the desired balance between enhanced efficiency and effective heat exchange.

3.5 Heat Losses Introduction Results

Table 3.5.1 presents the outcomes ensuing from the adjustment of parameters to reflect a more realistic operational scenario. These refined parameters, which include reduced pump efficiency [20] and heat exchanger efficiency set at 0.8 [21], contribute to a more accurate depiction of the system's performance under real-world conditions. The tabulated results serve as a valuable resource for evaluating the impact of these realistic considerations on various performance metrics within the model.

Table 3.3: Results for the final Selection of parameters after Taking into consideration the Heat Losses

T2 [°C]	T4 [°C]	Q23 [kW]	Q41 [kW]	Power Compressor [kW]	COP Heating	COP Cooling
107	-10	90.515	68.295	27.755	3.25	2.45

3.6 Solar – Thermal System Introduction

In the execution of this model, a pivotal consideration was the allocation of a total area of 10, 20 and 30 m^2 for the solar thermal system. This parameter played a crucial role in defining the capacity and effectiveness of the solar thermal component within the overall system. The Initial conditions are as mentioned before, thus mass flow rate of the refrigerant is set to 0.08 kg/s, maximum pressure to 1.5 MPa, initial pressure to 300 kPa and the refrigerant type being R717. The results are shown in table 3.4.

Table 3.4: Results after the Introduction of the Solar Thermal System

Area [m^2]	T2 [°C]	T4 [°C]	Q23 [kW]	Q41 [kW]	Power Compressor [kW]	COP Heating	COP Cooling
10	106	-10	83.982	68.295	24.511	3.42	2.78
20	106	-10	80.782	68.295	19.511	4.14	3.50
30	106	-10	77.582	68.295	14.511	5.34	4.70

3.7 Results for Water Heating and Cooling

The model designed for determining the mass flow rate of water, as supplied by the engineered heat pump, will incorporate equation 21 as a crucial element for the identification of this parameter. Equation 21 serves as a fundamental tool within the model, contributing to the comprehensive analysis aimed at accurately characterizing the heat pump system's performance and behavior. (Appendix V)

$$Q_{23} = M_o \, Cp_o \, dT_o \quad (21)$$

Where M_o is the mass of the fluid passing through the heat exchanger (refrigerant), Cp_o is the specific heat capacity of the fluid and dT_o is the temperature difference. In the case of equation 21, o is used to indicate the inner and outer fluid. While the equation for the heat provided to the outside fluid is shown in equation 22, [9]

$$Q_{23} = \dot{m}_{refrigerant} \, (h_2 - h_3) \quad (22)$$

The temperature of the fluid entering the heat exchanger is set to 25 °C while we want to achieve a temperature of 90 °C. For the calculation of the mass flow rates of water passing through the condenser and the evaporator, equations 23 and 24 are used, respectively [8]

$$\dot{m}_{refrigerant\ condeser} = \frac{Q_{23}}{C_{p,fluid}\,(T_{water_{out}} - (273.15+25))} \quad (23)$$

$$\dot{m}_{refrigerant\ evaporator} = \frac{Q_{41}}{C_{p,fluid}\,(T_{water_{out}} - (273.15+25))} \quad (24)$$

Using the values provided by the previous model for the R717 refrigerant using the solar thermal system with 30 m^2 of Area.

$$\dot{m}_{water\ condeser} = 0.2854 \pm \delta\dot{m}_{refrigerant\ condeser} \ kg/s$$

$$\dot{m}_{water\ evaporator} = 1.0889 \pm \delta\dot{m}_{refrigerant\ evaporator} \ kg/s$$

The relative errors for mass flow rate of water:

$$\delta\dot{m}_{refrigerant\ condeser} = \sqrt{(\frac{\partial\dot{m}}{\partial Q_{23}}\delta Q_{23})^2} = \sqrt{(\frac{0.2854}{77.582} \times 0.296)^2} = 0.001 \text{ kg/s}$$

$$\delta\dot{m}_{refrigerant\ evaporator} = \sqrt{(\frac{\partial\dot{m}}{\partial Q_{41}}\delta Q_{41})^2} = \sqrt{(\frac{1.0889}{68.295} \times 0.175)^2} = 0.0028 \text{ kg/s}$$

Chapter 4

4 MODEL VALIDATION

4.1 Model validation and error analysis

In the process of validating the model, a comparative analysis was undertaken by comparing the model's data against results from a published paper. The referenced paper involved a team that had previously conducted a comprehensive comparison of their model with experimental outcomes derived from laboratory testing conditions stipulated by ASHRAE [22].

To enhance the model's validation, a deliberate introduction of losses was incorporated into the heat exchanger. This was achieved by integrating a coefficient for losses (denoted as 0.8) into Equations 6 and 7. Additionally, an extra loss term was introduced for the compressor, aggregating the overall losses to 0.8. It is important to note that this approach diverged from the methodology employed in the referenced paper, where the team was using a dynamic model for the heat exchanger. It is noteworthy that the model was executed with a constant mass flow rate of 0.085 kg/s for the purpose of this validation.

For the calculations regarding the uncertainty of the model the equations for the absolute error (equation 21) as well as the equation for the Root Mean Square error (equation 26) will be used.

$$e_a = \frac{|\,Exp.-Simul.|}{Exp.} \times 100\% \quad (25)$$

$$RMS = \sqrt{\frac{\Sigma\,(x-x_i)^2}{N}} \quad (26)$$

Table 4.1: Experimental Results compared with the simulated results under test conditions A

	Test A		
	Experimental	**Simulation**	e_a **[%]**
Condenser [kW]	12.5	12.30	1.6
Evaporator [kW]	10.2	10.36	1.45
COP	4.0	4.05	1.25
Pmax [MPa]	2.6	2.6	
Pmin [MPa]	1.1	1.1	

Table 4.2: Experimental Results compared with the simulated results under test conditions B

	Test B		
	Experimental	**Simulation**	e_a **[%]**
Condenser [kW]	13.4	12.71	5.14
Evaporator [kW]	10.9	10.94	0.367
COP	4.9	5.02	2.45
Pmax [MPa]	2.26	2.26	
Pmin [MPa]	1.1	1.1	

Table 4.3: Experimental Results compared with the simulated results under test conditions C

	Test C		
	Experimental	**Simulation**	e_a **[%]**
Condenser [kW]	11.9	12.20	2.52
Evaporator [kW]	9.6	10.52	9.37
COP	4.5	4.64	3.11
Pmax [MPa]	2.2	2.2	
Pmin [MPa]	1.0	1.0	

Table 4.4: Average Values for Errors

	Experimental	**RMS error**	e_a **[%]**
Condenser [kW]	12.6	0.296	3.08
Evaporator [kW]	10.23	0.175	3.73
COP	4.467	0.412	2.27

The obtained results demonstrate a satisfactory alignment between the simulated and experimental data, revealing a strong correlation across all parameters. The assessment of uncertainty, detailed in Table 8, indicates generally low levels of uncertainty, underscoring the reliability of the model. Notably, in test C, there was a marginal increase in the uncertainty of heat exchange rates.

However, it is pertinent to note a consistent trend in the model's tendency to over-predict the Coefficient of Performance (COP). While the discrepancies were small in magnitude across all cases, this recurring pattern warrants further scrutiny and consideration in the ongoing refinement of the model. The overall agreement between the simulated and experimental data lays a solid foundation for the model's credibility, yet continuous efforts may be directed toward addressing and mitigating the observed over-prediction of COP values.

Chapter 5

5 CONCLUSIONS AND FUTURE WORK

5.1 Conclusions

This work proposes a high temperature heat pump that utilizes R717 as the working refrigerant, while also providing enough information for the design of the system, as well as further analysis. Detail analysis for all system parameters was carried out, concluding to the different ways each parameter can affect the system in a specific way. The parameters selected are 0.08 kg/s for the mass flow rate of the refrigerant, 1.5 MPa and 300 kPa for the maximum and initial pressure of the system.

The results confirm that the mass flow rate of the system will only increase the amount of heat provided by the pump, while also increase the amount of energy requires by the compressor, however it will not have any impact in the Coefficient of performance of the system neither in heating nor in cooling mode operation. While also showing that the initial pressure of the system has the significant role of improving the efficiency of the system, by an analogues trend, since a rise in the initial pressure of the system will result to an increase of the COP for both cooling and heating modes. Detail analysis for initial pressure results was carried out, while also being able to identify the effect of the initial pressure in all system energy transfer methods, as the heat exchange rate for both cooling and heating as well as the power required by the compressor.

From the Results regarding the refrigerants that were tested in the model created, the selection of R717 was clear, since it was superior to its competitors in all factors, having the highest coefficient of performance, while also achieving the highest temperatures compared to the standard refrigerants, such as R152 A. However, a reference for the R152 A as a possible refrigerant must be made since it is much safer than ammonia due to its chemical composition. Also, R152 A can result in quite effective outcome, as shown in table 3.1, since a 2.2 Coefficient of Performance for Heating was able to be calculated in the model.

Most important conclusion was the enormous increase in the efficiency of the system with the introduction of the solar thermal system in the cycle, that concluded to a very significant increase of the Coefficient of Performance of the system for both heating and cooling mode, while also give assistance in the effort of a more environmentally friendly system. Such an

introduction to the system will result in minimizing the electrical energy that must be supplied in the system using conventional energy methods and therefore making it a more sustainable system, being able to reach easier futures Europe's goals. However, the introduction of the Solar thermal system led to a slight decrease in the High temperature of the system, but still having a value over than 100 degrees Celsius and therefore, not being able to outweigh the advantage of this introduction. Even though this decreased occur, it must be mentioned that it is not a significant disadvantage, since the main purpose of the system is achievable as mentioned in Chapter 3.7.

Finally results for the heating of water from 25 to 100 °C are extremely promising, since the extracted results shows that approximately 285 grams of water can be heated to such temperatures every second, while the cooling reaches values of 1.09 kilograms per second. This is the most significant outcome of the research, showing the capabilities of this parameter selection, making the heat pump as efficient as possible, while also being able to achieve high amounts of water being able to heat at 100 degrees Celsius and making it able to be combined with several systems.

5.2 Future work

Concluding this study, is evident that significant avenues for refinement persist. After this analysis, imperative strides entail the formulation of a dynamic model capable of capturing the temporal evolution of heat pump parameters. Furthermore, such a model must possess enhanced predictive capabilities to temperatures changes across various nodes within the system with great precision. Unlike the existing model, which is constrained to furnishing outcomes solely for the primary four system's components, an advanced dynamic framework is poised to furnish a more comprehensive understanding of system dynamics and efficiency. Through the incorporation of dynamic elements and the expansion of analytical results, this forthcoming phase of investigation endeavors to refine prognostications and provide deeper insights into the intricate dynamics governing heat pump operations.

In the coming years, the development and manufacturing of an actual heat pump could serve as a pivotal step in validating and refining the theoretical model previously discussed. Such a practical endeavor promises not only to enhance the accuracy of our study but also to provide invaluable insights into the intricacies of the parameters discussed in earlier chapters. By subjecting the heat pump to real-world testing, we can gain a deeper understanding of its

performance under varying conditions, thus bolstering the credibility of our findings. Moreover, this hands-on approach equips future students with a comprehensive understanding of the factors that significantly influence the operation of such systems.

Furthermore, the realization of a full-scale heat pump model holds immense potential in terms of market competitiveness. As the demand for sustainable heating solutions continues to escalate, a well-designed heat pump could emerge as a compelling option, driving positive change in our transition to a more environmentally conscious future. It's worth noting that heating applications, particularly in the realm of building infrastructure, represent a cornerstone of global energy consumption. By offering a reliable and efficient heating solution, our project stands poised to revolutionize conventional approaches in an era of heightened energy efficiency and sustainability.

An additional idea will be the research of a two-stage cycle for the heat pump [23] and compare the efficiency and the applicability of such system, with the one stage model that has already been tested, using the same parameters that this paper has concluded to. Importantly enough, the assessment of pricing strategies concerning the systems involves analyzing both their cost of the system components and the appropriate pricing levels must be taken into consideration. This evaluation aims to strike a balance between the advantages gained from the systems and the financial outlay required to obtain them.

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Appendices

7 APPENDICES

Appendix I – MatLab Code for Base Model

```
clear,clc

%install python
pe=pyenv('Version','C:\Users\kotsi\AppData\Local\Programs\Python\Python311\python
.exe');
[v,e] = pyversion;
system([e,' -m pip install --user -U CoolProp']);

%% for loop to plot graphs
pressure=[]; entropy=[];enthalpy=[];Temperature=[]; initial_pressure_points=[];
COP_new=[]; p3_new=[];p2_new=[]; W_pump_points=[];
m_tonos=[]; Q_cooling=[]; Q=[]; p1_new=[]; COPc_new=[];
initial_pressure=50000;
for p1=200000:20000:200000
m_tonos_refrigerant_lower=0.08;
close all
%% Compressor
fluid='R717';
%Assumption of 4MPa
p2=1500000;
nps_compressor=0.85; %isentropic efficiency tou compressor
T1=py.CoolProp.CoolProp.PropsSI('T','Q',1,'P',p1,fluid);
fprintf('T1= %12.4f K \n',T1);
h1=py.CoolProp.CoolProp.PropsSI('H','Q',1,'P',p1,fluid);
s1=py.CoolProp.CoolProp.PropsSI('S','Q',1,'P',p1,fluid);
s2=s1; %isentropic
T2=py.CoolProp.CoolProp.PropsSI('T','P',p2,'S|gas',s2,fluid); %for isentropic
expansion
h2s=py.CoolProp.CoolProp.PropsSI('H','T',T2,'S|gas',s2,fluid);
Ws_pump=h2s-h1;
W_pump_per_kg=Ws_pump/nps_compressor;
h2=W_pump_per_kg+h1;
W_pump=W_pump_per_kg*m_tonos_refrigerant_lower;
Cp_refrigerant=py.CoolProp.CoolProp.PropsSI('C','Q',1,'P',p1,fluid);

fprintf('W_pump_per_kg= %12.4f J/kg \n', W_pump_per_kg);
fprintf('W_pump = %12.4f W \n', W_pump);

%% 1st Heat Exchanger
%Assumption of air temperature to be 25C
m_tonos_fluid=0.8; %(kg/s)
%also known that at point 6 quality must be 0

p3=p2;
s3=py.CoolProp.CoolProp.PropsSI('S','Q',0,'P',p3,fluid);
h3=py.CoolProp.CoolProp.PropsSI('H','Q',0,'P',p3,fluid);
T3=py.CoolProp.CoolProp.PropsSI('T','Q',0,'P',p3,fluid);
Q23=m_tonos_refrigerant_lower*(h2-h3);

Cp_air=py.CoolProp.CoolProp.PropsSI('C','T',273+25,'P',101325,'AIR');
Q_water=((m_tonos_fluid*Cp_air)*((273.15+100)-(273.15+25)));
```

```
%xerontas auto mporo na rithmiso tin thermokrasia stin opia tha vgeni ton nero i
tin roi tou nerou

fprintf('T3= %12.4f K \n',T3);
fprintf('Q23= %12.4f W \n',Q23);

%% expansio valve 3-4 heat exchanger 4-1

%isenthalpic

h4=h3;
p4=p1;
x4=py.CoolProp.CoolProp.PropsSI('Q','H',h4,'P',p4,fluid);
s4=py.CoolProp.CoolProp.PropsSI('S','H',h4,'P',p4,fluid);
T4=py.CoolProp.CoolProp.PropsSI('T','P',p4,'H',h4,fluid);
fprintf('T4= %12.4f K \n', T4);
fprintf('x4= %12.4f \n',x4);
Q41= m_tonos_refrigerant_lower*abs((h4-h1));
COPc=Q41/W_pump;


%% Calculation of COP
% Check for validity
if p3>p4
COP=Q23/(W_pump);
fprintf('COP= %12.4f \n',COP);
COP_new=[COP_new, COP];
COPc_new=[COPc_new,COPc];

%% everything to matrices
entropy=[entropy;[s1,s2,s3,s4,s1]];
pressure=[pressure;[p1,p2,p3,p4,p1]];
enthalpy=[enthalpy;[h1,h2,h3,h4,h1]];
Temperature=[Temperature;[T1,T2,T3,T4,T1]];
m_tonos=[m_tonos;m_tonos_refrigerant_lower];

W_pump_points=[W_pump_points,W_pump];
initial_pressure_points=[initial_pressure_points;p1];
p3_new=[p3_new,p3];
p2_new=[p2_new,p2];
p1_new=[p1_new,p1];
Q=[Q;Q23];
Q_cooling=[Q_cooling;Q41];
else
    continue
end
%% get ou if negative Q
if Q23<0
    break
end
end

%FIT DATA
cooling_poly=polyfit(p1_new,Q_cooling,2);
heating_poly=polyfit(p1_new,Q,2);
pump_poly=polyfit(p1_new,W_pump_points,2);

Q_cooling_poly=polyval(cooling_poly,10000:1000:200000);
Q_heating_poly=polyval(heating_poly,10000:1000:200000);
P_pump_poly=polyval(pump_poly,10000:1000:200000);
```

```
%% Thermodynamic Diagrams
P_fluid=[];T_gas=[];T_fluid=[];H_gas=[];H_fluid=[];S_gas=[];S_fluid=[];
for i=initial_pressure/3:initial_pressure/100:initial_pressure*100
    saturated=py.CoolProp.CoolProp.PropsSI('S','P',i,'Q',1,fluid);
    if saturated==-1
        continue
    end
    S_fluid=[S_fluid,py.CoolProp.CoolProp.PropsSI('S','P',i,'Q',0,fluid)];
    T_fluid=[T_fluid,py.CoolProp.CoolProp.PropsSI('T','P',i,'Q',0,fluid)];
    H_fluid=[H_fluid,py.CoolProp.CoolProp.PropsSI('H','P',i,'Q',0,fluid)];
    S_gas=[S_gas,py.CoolProp.CoolProp.PropsSI('S','P',i,'Q',1,fluid)];
    T_gas=[T_gas,py.CoolProp.CoolProp.PropsSI('T','P',i,'Q',1,fluid)];
    H_gas=[H_gas,py.CoolProp.CoolProp.PropsSI('H','P',i,'Q',1,fluid)];
    P_fluid=[P_fluid,i];
end

%% Plot after the error
S_gas_new=[];H_gas_new=[];T_gas_new=[];P_gas_new=[];
for i=1:length(P_fluid)
    S_gas_new=[S_gas_new,S_gas(length(P_fluid)-i+1)];
    H_gas_new=[H_gas_new,H_gas(length(P_fluid)-i+1)];
    T_gas_new=[T_gas_new,T_gas(length(P_fluid)-i+1)];
    P_gas_new=[P_gas_new,P_fluid(length(P_fluid)-i+1)];
end
S=[S_fluid,S_gas_new]; H=[H_fluid,H_gas_new]; T=[T_fluid,T_gas_new];
P=[P_fluid,P_gas_new];
%% cycles
str_new=[];

hold on
figure(1)
for i=1:length(pressure(:,1))
    plot(enthalpy(i,:),pressure(i,:))
end
plot(H,P,'-r')
title('Enthalpy versus Pressure Graph')
xlabel('Enthalpy [J/kg]')
ylabel('Pressure [Pa]')
grid on
hold off


hold on
figure(2)
for i=1:length(pressure(:,1))
    plot(entropy(i,:),pressure(i,:))
end
plot(S,P,'r')
title('Entropy versus Pressure Graph')
xlabel('Entropy [J/kg K]')
ylabel('Pressure [Pa]')
grid on
hold off

hold on
figure(3)
str_new=[];
for i=1:length(pressure(:,1))
    plot(entropy(i,:),Temperature(i,:))
end
```

```
plot(S,T,'r')
title('Entropy versus Temperature Graph')
xlabel('Entropy [J/kg K]')
ylabel('Temperature [K]')
grid on
hold off

figure(4)
plot(p1_new,COP_new)
xlabel('Initial Pressure [Pa]'), ylabel('COP')
grid on

figure(5)
plot(p1_new,Q)
xlabel('Initial Pressure [Pa]'), ylabel('Rate of Heat Exchanged [W]')
grid on

figure(6)
plot(p1_new,Q_cooling)
xlabel('Initial Pressure [Pa]'), ylabel('Rate of Heat Exchanged cooling mode
[W]')
grid on

figure(7)
plot(p1_new,W_pump_points)
xlabel('Initial Pressure [Pa]'), ylabel('Power of the Pump [W]')
grid on

figure(8)
plot(p1_new,COPc_new)
xlabel('Initial Pressure [Pa]'), ylabel('COP cooling')
grid on

%%poly values

figure(9)
plot(10000:1000:200000,Q_heating_poly)
xlabel('Initial Pressure [Pa]'), ylabel('Rate of Heat Exchanged [W]')
grid on

figure(10)
plot(10000:1000:200000,Q_cooling_poly)
xlabel('Initial Pressure [Pa]'), ylabel('Rate of Heat Exchanged cooling mode
[W]')
grid on

figure(11)
plot(10000:1000:200000,P_pump_poly)
xlabel('Initial Pressure [Pa]'), ylabel('Power of the Pump [W]')
grid on
```

Appendix II – Mass Flow Rate, Initial Pressure Analysis and Refrigerant Selection

The Analysis for the mass flow rate and initial pressure and the selection of the refrigerant was based on the code in Appendix I with the addition of the extra coding shown in Appendix II

```
%% Thermodynamic Diagrams
P_fluid=[];T_gas=[];T_fluid=[];H_gas=[];H_fluid=[];S_gas=[];S_fluid=[];
for i=initial_pressure/3:initial_pressure/100:initial_pressure*100
    saturated=py.CoolProp.CoolProp.PropsSI('S','P',i,'Q',1,fluid);
    if saturated==-1
        continue
    end
    S_fluid=[S_fluid,py.CoolProp.CoolProp.PropsSI('S','P',i,'Q',0,fluid)];
    T_fluid=[T_fluid,py.CoolProp.CoolProp.PropsSI('T','P',i,'Q',0,fluid)];
    H_fluid=[H_fluid,py.CoolProp.CoolProp.PropsSI('H','P',i,'Q',0,fluid)];
    S_gas=[S_gas,py.CoolProp.CoolProp.PropsSI('S','P',i,'Q',1,fluid)];
    T_gas=[T_gas,py.CoolProp.CoolProp.PropsSI('T','P',i,'Q',1,fluid)];
    H_gas=[H_gas,py.CoolProp.CoolProp.PropsSI('H','P',i,'Q',1,fluid)];
    P_fluid=[P_fluid,i];
end
```

```
%% Plot after the error
S_gas_new=[];H_gas_new=[];T_gas_new=[];P_gas_new=[];
for i=1:length(P_fluid)
    S_gas_new=[S_gas_new,S_gas(length(P_fluid)-i+1)];
    H_gas_new=[H_gas_new,H_gas(length(P_fluid)-i+1)];
    T_gas_new=[T_gas_new,T_gas(length(P_fluid)-i+1)];
    P_gas_new=[P_gas_new,P_fluid(length(P_fluid)-i+1)];
end
S=[S_fluid,S_gas_new]; H=[H_fluid,H_gas_new]; T=[T_fluid,T_gas_new]; P=[P_fluid,P_gas_new];
```

```
%% cycles
str_new=[];
hold on
figure(1)
for i=1:length(pressure(:,1))
    plot(enthalpy(i,:),pressure(i,:))
    % str=num2str(initial_pressure_points(i));
    % str_new=[str_new,',', str];
    % legend(str_new,'Location','northwest','NumColumns',i)
end
plot(H,P,'-r')
title('Enthalpy versus Pressure Graph')
xlabel('Enthalpy [J/kg]')
ylabel('Pressure [Pa]')
grid on
hold off
```

Appendix III – Analysis Regarding Heat Losses

```
clear,clc

%install python
pe=pyenv('Version','C:\Users\kotsi\AppData\Local\Programs\Python\Python311\python
.exe');
[v,e] = pyversion;
system([e,' -m pip install --user -U CoolProp']);

%% for loop to plot graphs
pressure=[]; entropy=[];enthalpy=[];Temperature=[]; initial_pressure_points=[];
COP_new=[]; p3_new=[];p2_new=[]; W_pump_points=[];
m_tonos=[]; Q_cooling=[]; Q=[]; p1_new=[]; COPc_new=[];
initial_pressure=200000;
for p1=300000:50000:300000
m_tonos_refrigerant_lower=0.08;
close all
%% Compressor
fluid='R717';
%Assumption of 4MPa
p2=1500000;
nps_compressor=0.85; %isentropiki apodotikotia tou compressor
T1=py.CoolProp.CoolProp.PropsSI('T','Q',1,'P',p1,fluid);
fprintf('T1= %12.4f K \n',T1);
h1=py.CoolProp.CoolProp.PropsSI('H','Q',1,'P',p1,fluid);
s1=py.CoolProp.CoolProp.PropsSI('S','Q',1,'P',p1,fluid);
s2=s1; %isentropic
T2=py.CoolProp.CoolProp.PropsSI('T','P',p2,'S|gas',s2,fluid); %for isentropic
expansion
h2s=py.CoolProp.CoolProp.PropsSI('H','T',T2,'S|gas',s2,fluid);
Ws_pump=h2s-h1;
W_pump_per_kg=(Ws_pump/nps_compressor)/0.8;
h2=W_pump_per_kg+h1;
W_pump=W_pump_per_kg*m_tonos_refrigerant_lower;
Cp_refrigerant=py.CoolProp.CoolProp.PropsSI('C','Q',1,'P',p1,fluid);

fprintf('W_pump_per_kg= %12.4f J/kg \n', W_pump_per_kg);
fprintf('W_pump = %12.4f W \n', W_pump);

%% 3rd Heat Exchanger
%Assumption of air temperature to be 25C
m_tonos_fluid=0.6; %(kg/s)
%also known that at point 6 quality must be 0

p3=p2;
s3=py.CoolProp.CoolProp.PropsSI('S','Q',0,'P',p3,fluid);
h3=py.CoolProp.CoolProp.PropsSI('H','Q',0,'P',p3,fluid);
T3=py.CoolProp.CoolProp.PropsSI('T','Q',0,'P',p3,fluid);
Q23=m_tonos_refrigerant_lower*(h2-h3)*0.8;

Cp_air=py.CoolProp.CoolProp.PropsSI('C','T',273+25,'P',101325,'AIR');
Q_water=((m_tonos_fluid*Cp_air)*((273.15+100)-(273.15+25)));

fprintf('T3= %12.4f K \n',T3);
fprintf('Q23= %12.4f W \n',Q23);

%% expansio valve 7-8
```

```
%isenthalpic

h4=h3;
p4=p1;
x4=py.CoolProp.CoolProp.PropsSI('Q','H',h4,'P',p4,fluid);
s4=py.CoolProp.CoolProp.PropsSI('S','H',h4,'P',p4,fluid);
T4=py.CoolProp.CoolProp.PropsSI('T','P',p4,'H',h4,fluid);
fprintf('T4= %12.4f K \n', T4);
fprintf('x4= %12.4f \n',x4);
Q41= m_tonos_refrigerant_lower*abs((h4-h1))*0.8;
COPc=Q41/W_pump;


%% Calculation of COP
% Check for validity
if p3>p4
COP=Q23/(W_pump);
fprintf('COP= %12.4f \n',COP);
COP_new=[COP_new, COP];
COPc_new=[COPc_new,COPc];

%% everything to matrices
entropy=[entropy;[s1,s2,s3,s4,s1]];
pressure=[pressure;[p1,p2,p3,p4,p1]];
enthalpy=[enthalpy;[h1,h2,h3,h4,h1]];
Temperature=[Temperature;[T1,T2,T3,T4,T1]];
m_tonos=[m_tonos;m_tonos_refrigerant_lower];

W_pump_points=[W_pump_points,W_pump];
initial_pressure_points=[initial_pressure_points;p1];
p3_new=[p3_new,p3];
p2_new=[p2_new,p2];
p1_new=[p1_new,p1];
Q=[Q;Q23];
Q_cooling=[Q_cooling;Q41];
else
    continue
end
%% get ou if negative Q
if Q23<0
    break
end
end

%FIT DATA
cooling_poly=polyfit(p1_new,Q_cooling,2);
heating_poly=polyfit(p1_new,Q,2);
pump_poly=polyfit(p1_new,W_pump_points,2);

Q_cooling_poly=polyval(cooling_poly,10000:1000:200000);
Q_heating_poly=polyval(heating_poly,10000:1000:200000);
P_pump_poly=polyval(pump_poly,10000:1000:200000);


%% Thermodynamic Diagrams
P_fluid=[];T_gas=[];T_fluid=[];H_gas=[];H_fluid=[];S_gas=[];S_fluid=[];
for i=initial_pressure/3:initial_pressure/100:initial_pressure*100
    saturated=py.CoolProp.CoolProp.PropsSI('S','P',i,'Q',1,fluid);
    if saturated==-1
        continue
```

```
    end
    S_fluid=[S_fluid,py.CoolProp.CoolProp.PropsSI('S','P',i,'Q',0,fluid)];
    T_fluid=[T_fluid,py.CoolProp.CoolProp.PropsSI('T','P',i,'Q',0,fluid)];
    H_fluid=[H_fluid,py.CoolProp.CoolProp.PropsSI('H','P',i,'Q',0,fluid)];
    S_gas=[S_gas,py.CoolProp.CoolProp.PropsSI('S','P',i,'Q',1,fluid)];
    T_gas=[T_gas,py.CoolProp.CoolProp.PropsSI('T','P',i,'Q',1,fluid)];
    H_gas=[H_gas,py.CoolProp.CoolProp.PropsSI('H','P',i,'Q',1,fluid)];
    P_fluid=[P_fluid,i];
end

%% Plot after the error
S_gas_new=[];H_gas_new=[];T_gas_new=[];P_gas_new=[];
for i=1:length(P_fluid)
    S_gas_new=[S_gas_new,S_gas(length(P_fluid)-i+1)];
    H_gas_new=[H_gas_new,H_gas(length(P_fluid)-i+1)];
    T_gas_new=[T_gas_new,T_gas(length(P_fluid)-i+1)];
    P_gas_new=[P_gas_new,P_fluid(length(P_fluid)-i+1)];
end
S=[S_fluid,S_gas_new]; H=[H_fluid,H_gas_new]; T=[T_fluid,T_gas_new];
P=[P_fluid,P_gas_new];
%% cycles
str_new=[];

hold on
figure(1)
for i=1:length(pressure(:,1))
    plot(enthalpy(i,:),pressure(i,:))
end
plot(H,P,'-r')
title('Enthalpy versus Pressure Graph')
xlabel('Enthalpy [J/kg]')
ylabel('Pressure [Pa]')
grid on
hold off


hold on
figure(2)
for i=1:length(pressure(:,1))
    plot(entropy(i,:),pressure(i,:))
end
plot(S,P,'r')
title('Entropy versus Pressure Graph')
xlabel('Entropy [J/kg K]')
ylabel('Pressure [Pa]')
grid on
hold off

hold on
figure(3)
str_new=[];
for i=1:length(pressure(:,1))
    plot(entropy(i,:),Temperature(i,:))
    %str=num2str(initial_pressure_points(i));
    %str_new=[str_new; str];
    %legend(str_new,'Location','northwest','NumColumns',i)
end
plot(S,T,'r')
title('Entropy versus Temperature Graph')
xlabel('Entropy [J/kg K]')
ylabel('Temperature [K]')
```

```
grid on
hold off

figure(4)
plot(p1_new,COP_new)
xlabel('Initial Pressure [Pa]'), ylabel('COP')
grid on

figure(5)
plot(p1_new,Q)
xlabel('Initial Pressure [Pa]'), ylabel('Rate of Heat Exchanged [W]')
grid on

figure(6)
plot(p1_new,Q_cooling)
xlabel('Initial Pressure [Pa]'), ylabel('Rate of Heat Exchanged cooling mode
[W]')
grid on

figure(7)
plot(p1_new,W_pump_points)
xlabel('Initial Pressure [Pa]'), ylabel('Power of the Pump [W]')
grid on

figure(8)
plot(p1_new,COPc_new)
xlabel('Initial Pressure [Pa]'), ylabel('COP cooling')
grid on

%%poly values

figure(9)
plot(10000:1000:200000,Q_heating_poly)
xlabel('Initial Pressure [Pa]'), ylabel('Rate of Heat Exchanged [W]')
grid on

figure(10)
plot(10000:1000:200000,Q_cooling_poly)
xlabel('Initial Pressure [Pa]'), ylabel('Rate of Heat Exchanged cooling mode
[W]')
grid on

figure(11)
plot(10000:1000:200000,P_pump_poly)
xlabel('Initial Pressure [Pa]'), ylabel('Power of the Pump [W]')
grid on
```

Appendix IV – Introduction of a Solar Thermal System

The solar thermal model is the same as the refrigerant model, with the difference in the compressor part shown below.

```
%% Compressor
fluid='R717';
%Assumption of 4MPa
p2=1500000;
nps_compressor=0.8; %isentropiki apodotikotia tou compressor
T1=py.CoolProp.CoolProp.PropsSI('T','Q',1,'P',p1,fluid);
fprintf('T1= %12.4f K \n',T1);
h1=py.CoolProp.CoolProp.PropsSI('H','Q',1,'P',p1,fluid);
s1=py.CoolProp.CoolProp.PropsSI('S','Q',1,'P',p1,fluid);

s_endiameso=s1;
Q=(800*0.40*30)/m_tonos_refrigerant_lower;
h_endiameso=h1+Q;
%p_endiameso=py.CoolProp.CoolProp.PropsSI('P','H',h_endiameso,'S',s_endiameso,flu
id);
%T_endiameso=py.CoolProp.CoolProp.PropsSI('T','H',h_endiameso,'S',s_endiameso,flu
id);

s2=s_endiameso; %isentropic

%isentropic
T2=py.CoolProp.CoolProp.PropsSI('T','P',p2,'S|gas',s2,fluid); %for isentropic
expansion
h2s=py.CoolProp.CoolProp.PropsSI('H','T',T2,'S|gas',s2,fluid);

Ws_pump=h2s-h_endiameso;
W_pump_per_kg=Ws_pump/nps_compressor;
h2=W_pump_per_kg+h1;
W_pump=(W_pump_per_kg*m_tonos_refrigerant_lower)/0.8;
Cp_refrigerant=py.CoolProp.CoolProp.PropsSI('C','Q',1,'P',p1,fluid);

fprintf('W_pump_per_kg= %12.4f J/kg \n', W_pump_per_kg);
fprintf('W_pump = %12.4f W \n', W_pump);
```

Appendix V - Water Heating and Cooling

The Water heating and cooling mode is the same as the basic model with the difference being at the heat exchangers

```
%% 1st Heat Exchanger
%Assumption of air temperature to be 25C %(kg/s)
%also known that at point 6 quality must be 0

p3=p2;
s3=py.CoolProp.CoolProp.PropsSI('S','Q',0,'P',p3,fluid);
h3=py.CoolProp.CoolProp.PropsSI('H','Q',0,'P',p3,fluid);
T3=py.CoolProp.CoolProp.PropsSI('T','Q',0,'P',p3,fluid);
Q23=m_tonos_refrigerant_lower*(h2-h3)*0.8;

Cp_water=py.CoolProp.CoolProp.PropsSI('C','T',273+25,'P',101325,'Water');
M_tonos_water_cond=Q23/((Cp_water)*((273.15+90)-(273.15+25)));

fprintf('T3= %12.4f K \n',T3);
fprintf('Q23= %12.4f W \n',Q23);

%% expansio valve 3-4 Heat Exchanger 4-1

%isenthalpic

h4=h3;
p4=p1;
x4=py.CoolProp.CoolProp.PropsSI('Q','H',h4,'P',p4,fluid);
s4=py.CoolProp.CoolProp.PropsSI('S','H',h4,'P',p4,fluid);
T4=py.CoolProp.CoolProp.PropsSI('T','P',p4,'H',h4,fluid);
fprintf('T4= %12.4f K \n', T4);
fprintf('x4= %12.4f \n',x4);
Q41= m_tonos_refrigerant_lower*abs((h4-h1))*0.8;
COPc=Q41/W_pump;
M_tonos_water_evap=Q41/((Cp_water)*((273.15-10)-(273.15+25)));
```